



Experiment 2

Experimental Report on Intrinsic Calibration of Mobile Phone Camera

Individual Experiment Report

Author: Dong Xiaorui

Student ID: 24020036015

Group: 5

Date: 2026.9.26

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General Overview

This experiment focuses on the end-to-end integration of perception, task planning and real-robot execution for desktop object sorting tasks. Building upon previous simulation development for visual detection and robotic-arm control, this part migrates the algorithm framework from simulation to the physical mechArm robotic-arm hardware.

The whole system adopts a three-layer architecture including vision layer, task-control layer and execution layer. With the RGB overhead camera providing object perception, the ROS 2 framework mediates message passing between modules. The system recognizes square and circular objects, maps detected targets to pre-defined desktop grid positions, and drives the physical robotic arm to perform pick-and-place operations into corresponding classification bins.

Besides normal multi-object sequential classification, the experiment also considers practical hardware-oriented exceptions: empty grids, unrecognized objects and grasping failures. Safety verification for joint limits, collision avoidance and home-position reset are essential for real-robot deployment. Test results reflect real-world errors including calibration bias and gripper instability, which do not appear in simulation. This work lays the hardware foundation for further desktop-object intelligent manipulation.

1 Experimental Objectives

Building upon prior work in visual inspection, task control, and the development of a robotic arm simulation system, this experiment further completes the integrated system debugging of a real robotic arm platform. The experiment connects a real RGB camera, a ROS 2 communication system, a mechArm robotic arm, a gripper, and a task control module, enabling the system to perform grasping, transporting, and categorized placement of target objects based on visual inspection results.

- Establish and initialize communication between the ROS 2 system and the physical mechArm robotic arm. Verify the basic motion capabilities, gripper control, and safe HOME position of the physical robotic arm.
- Establish the correspondence between the physical camera field of view, the tabletop grid, and the fixed grasping positions of the robotic arm.
- Transmit visual detection results to the task control module to determine target grids and classification bins.
- Implement automatic grasping, classification, and placement of target objects by the physical robotic arm.
- Validate a multi-target sequential classification pipeline, and handle edge cases such as empty grids, unrecognized objects, and grasping failures.
- Ensure the operational safety and system stability of the physical robotic arm throughout its operation.

2 Experimental System and Overall Architecture

The physical hardware comprises an RGB camera mounted from above, a mechArm robotic arm, a gripper, a ROS2 master computer, an experimental workstation, target objects, and a classification bin. The software employs ROS2 as its communication and scheduling framework and includes modules for image acquisition, object detection, grid-based localization, classification logic, task management, and control of the robotic arm and gripper.

The system is structured into three layers:

- **Vision layer:** determines the position and category of objects;
- **Task control layer:** handles grid-based positioning and bin allocation;
- **Robotic arm execution layer:** converts task instructions into actual motion commands for the physical hardware.

The complete data flow is as follows: real-time RGB camera $\rightarrow$ object detection topic /detections $\rightarrow$ grid-based localization $\rightarrow$ category determination $\rightarrow$ bin selection $\rightarrow$ task dispatch /pick_and_place command $\rightarrow$ mechArm execution (grasping, transporting, and placement) $\rightarrow$ status feedback $\rightarrow$ loop back to process the next target.

In this experiment, we maximized the reuse of task interfaces and logic from the simulation phase, replacing only the simulation-based robotic arm execution unit with the physical mechArm control module, thereby reducing the effort required for the migration from simulation to physical implementation.

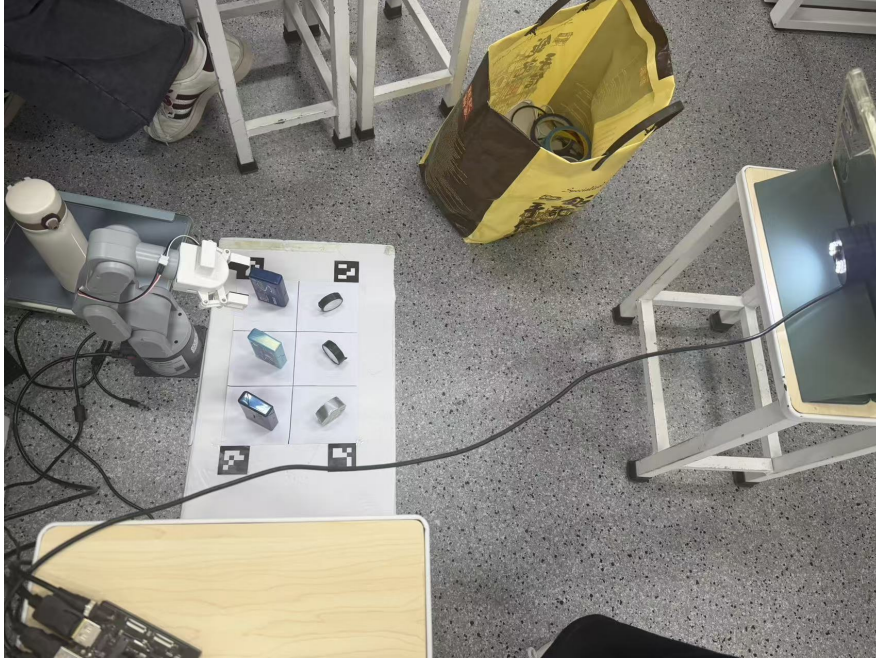


Figure 1. Overall architecture of automatic desktop-object classification system

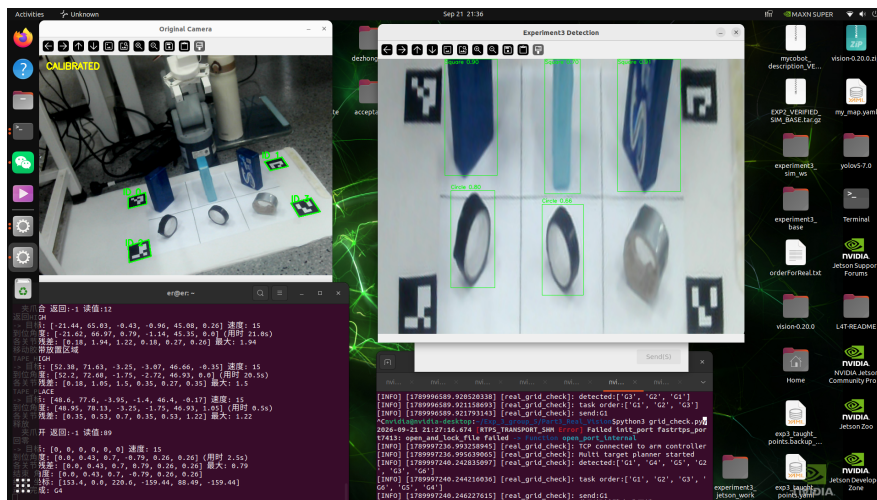


Figure 2. Running status of ROS2 nodes, topics and task interfaces

3 Physical Device Debugging, Grid Configuration and Fixed-Point Gripping Scheme

Prior to officially executing the automated classification task, complete the initialization of the mechArm physical robot connection, verify communication status, joint positions, and gripper states, and confirm the HOME safety position. Systematically debug each gripping point and each material bin placement point to ensure that all points lie within the robot’s workspace, and mitigate the risk of motion collisions.

The experimental workspace is divided into six picking grids (G1–G6), with two classification zones designated as Bin1 and Bin2. After camera calibration, establish a mapping between image pixel coordinates and the six grids, so that visual recognition of targets can be accurately mapped to the robot’s fixed gripping coordinates.

The experiment employs a **fixed-point gripping strategy**: once the task module identifies the target grid, it directly invokes the preconfigured gripping coordinates for that grid. A complete single operation cycle consists of: approaching the target, descending, closing the gripper, lifting, moving to the corresponding material bin, opening the gripper to place the object, retracting, and returning to the HOME safety position. This fixed-point approach reduces the burden of real-time motion planning, facilitating point-by-point debugging and safety verification.

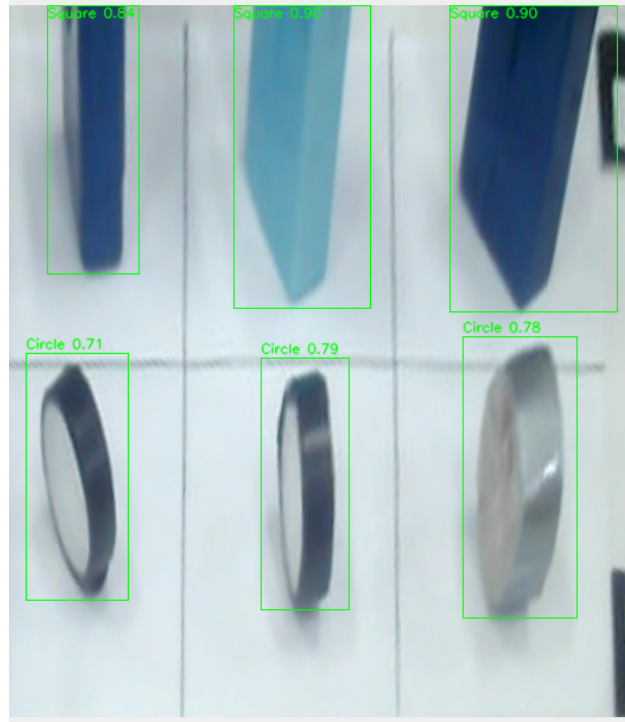


Figure 3. Desktop grids and classification regions under camera view

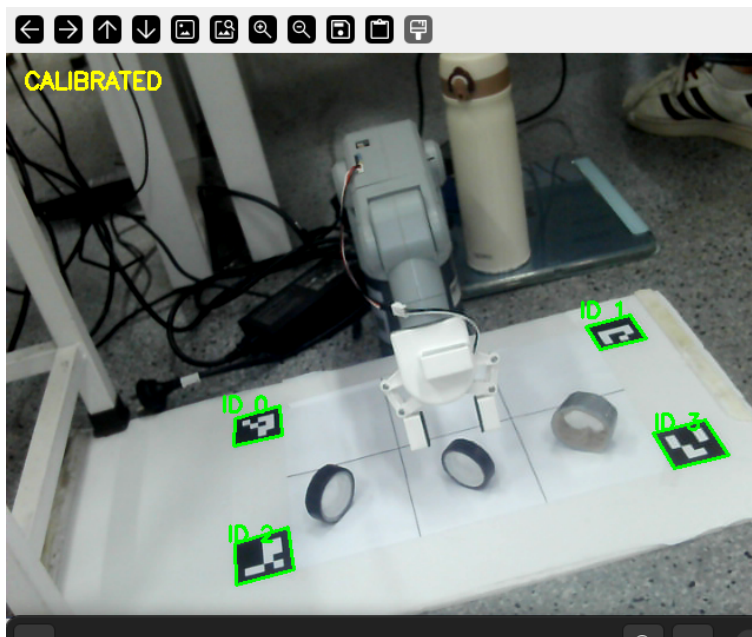


Figure 4. Illustration of six fixed grasping positions of mechArm

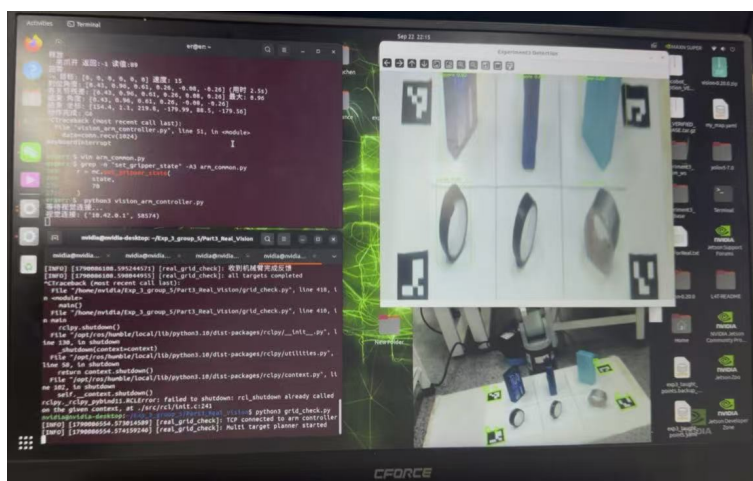


Figure 5. Physical grasping and classification placement process of mechArm

4 Complete Automated Classification Workflow

1. **Image Acquisition:** An RGB camera captures the desktop environment.
2. **Object Detection:** The vision module identifies objects and outputs their category and positional information.
3. **Grid and Category Classification:** Based on the center of the detection bounding box and calibration results, the system determines the corresponding grid and distinguishes between square and circular objects.
4. **Bin Assignment:** Square objects are routed to Bin1, while circular objects are routed to Bin2.
5. **Task Dispatch:** The task module packages the `grid_id` and `bin_id` to dispatch the grasping task.
6. **Robotic Arm Execution:** The arm moves to the designated grid to perform grasping, then transports the object to the material bin for placement.
7. **Reset and Looping:** Upon completion of the operation, the arm returns to its safe position and proceeds to process the next target until the task is finished.

5 Multi-object Automatic Classification Experiment and Exception Handling

5.1 Multi-Object Automatic Classification Experiment

To verify the full operational capability of the system, multiple classes of target objects are placed on a real desktop environment, and automatic classification is performed using visual detection and a task control system. A typical task configuration is shown in Table 1.

Table 1. Task configuration of multi-grid classification

Grid	Target Class	Target Bin
G1	Square	Bin 1
G2	Square	Bin 1
G3	Square	Bin 1
G4	Circle	Bin 2
G5	Circle	Bin 2
G6	Circle	Bin 2

Before initiating the experiment, the initial positions of the six target objects and their corresponding grids are first verified. Subsequently, the automated classification process is

launched, during which the system sequentially performs target recognition, grid matching, bin selection, robotic arm grasping, and placement.

5.2 Handling of Abnormal Conditions

[noitemsep]

- **Empty Grid:** When no target object is present within a given Grid, the visual detection module fails to acquire a valid target, and the task control module shall not issue a grasping command to the robotic arm. The system shall skip this Grid and proceed to inspect subsequent Grids, thereby preventing the robotic arm from executing futile empty grasping motions.
- **Unidentified Object:** If a target is detected but its category cannot be reliably determined, the system shall not immediately execute classification actions. For such targets, the system may either skip the target or enter a waiting state, avoiding placement of the object into the incorrect classification bin.
- **Grasping Failure:** If the robotic arm fails to successfully grasp the target, or if the target falls during handling, the current task shall be marked as failed, and all subsequent actions dependent on that target shall be halted. The robotic arm shall then return to a safe position or remain stationary in accordance with the safety protocol, preventing the execution of erroneous consecutive motions.

6 Safety Analysis

Real robotic arms impose stricter safety requirements compared to simulated environments. Therefore, prior to executing automated classification tasks, the following verifications must be performed:

[noitemsep]

1. Verify all fixed gripping positions, classification bin positions, and HOME positions.
2. Ensure that the robotic arm will not collide with the worktable during its motion.
3. Ensure that the robotic arm will not collide with the camera, its mounting bracket, or other experimental equipment.
4. Ensure that the classification bins are located within the reachable workspace of the robotic arm.
5. Ensure that the joint motion ranges of the robotic arm remain within safe limits.
6. Ensure that the gripper can safely perform closing and opening motions.

7. The robotic arm can be promptly stopped in the event of an abnormal condition.
8. Upon task completion, the robotic arm can return to its preset HOME or safe position.

7 Recording of Experimental Results

Upon completion of the actual hardware-based experiment, fill in the table according to on-site operation results. The number of successful trials, number of anomalies, and success rate shall be based on the actual experimental records.

Table 2. Experimental result record table

No.	Grid	Classification	Target Bin	Grasping Result	Placement Result	Remarks
1	G1	Square	Bin 1	success	success	
2	G2	Square	Bin 1	success	success	
3	G3	Square	Bin 1	success	success	
4	G4	Circle	Bin 2	success	success	
5	G5	Circle	Bin 2	success	success	
6	G6	Circle	Bin 2	Lose	Lose	Grasp-drop failure

- Total number of target objects: 6
- Number of successfully grasped objects: 5
- Number of correctly classified objects: 5
- Classification success rate: 100%
- Number of abnormal tasks: 1
- Number of collisions: 1
- Number of joint overload incidents: 0

8 Analysis of Experimental Results

This experiment successfully accomplished integrated commissioning among the visual inspection system, the ROS 2 task control system, and the physical mechArm robotic arm. Compared to the simulation environment, the physical system is subject to various factors—including camera installation errors, desktop coordinate calibration errors, robotic arm positioning errors, changes in target object positions, and gripper grasping stability—thereby precluding the direct application of physical coordinates from the simulation environment to the real hardware.

During the system migration process, maintaining consistency in the preliminary task interfaces and control logic can minimize the amount of modifications required when transferring the simulation system to the physical robotic arm. In this setup, the visual module continues to handle target detection, the task control module handles Grid and Bin classification, while the physical robotic arm control module manages specific motion execution and gripper actuation.

The key to successful integrated commissioning of the physical system lies in establishing reliable correspondences between image coordinates, desktop Grid coordinates, and robotic arm grasping coordinates, as well as conducting independent tests on each fixed grasping point. Only by ensuring safe and stable single-point operation can the system proceed to multi-target continuous automatic classification.

9 Experimental Summary

This experiment focused on the automated classification and organization task for desktop objects, achieving comprehensive integration and debugging among a real-world mechArm robotic arm, visual detection systems, and the ROS 2 task control system. The system is capable of determining the position and category of target objects based on visual detection results, subsequently selecting the appropriate classification bin, and then having the real robotic arm execute grasping, transport, and placement tasks.

This experiment demonstrates that when migrating from a simulation environment to a real robotic platform, the system must address practical challenges such as camera calibration, grid-based positioning, robotic arm positioning accuracy, gripper grasping stability, and handling of abnormal operating conditions. By employing strategies—including fixing grasping points, standardizing ROS 2 task interfaces, and defining safe HOME positions—the system’s debuggability and operational safety can be significantly enhanced. This phase has accomplished the system integration from a simulated robotic arm to a real-world mechArm robotic arm, thereby establishing a robust real robotic platform foundation for the subsequent completion of the full automated classification and organization task for desktop objects.